

Empirical Proof Record: VALIDATED

2026-05-24T03:38:21.622288+00:00

Empirical Proof Record — VALIDATED

Proof ID: 34cbb086f141f87867ec675989905560...

Created: 2026-05-24T03:38:21.622288+00:00

1. Claim

Across up to 5000 real FRED numbers within $\pm 1e+09$, the live number sense emits a monotone number line: $\text{Spearman}(\text{value}, \text{p_thermo}) \geq 0.90$.

- **Operationalisation:** Distinct real FRED values within range; live ArachneProtocol $\text{assign}(\text{str}(v)) \rightarrow \text{p_thermo}$; $\text{Spearman}(\text{value}, \text{p_thermo})$; distinctness; neighbour-locality (adjacent-rank prime gap vs random-pair gap); $\text{zero} \rightarrow \text{assign}('0').\text{p_thermo}$.
- **Threshold:** $\text{number_line_spearman} \geq 0.9$ rank-correlation
- **H0:** $\text{Spearman} < 0.90$ — the number sense does not preserve magnitude on real data
- **H1:** $\text{Spearman} \geq 0.90$ — a real magnitude number line

2. Verdict

VALIDATED — observed 0.9727888184955529

Spearman 0.9728 over 5000 real numbers $[-18.28, 3.40015e+07]$; distinct 0.669; neighbour-locality adj/random 0.0162; zero→1409

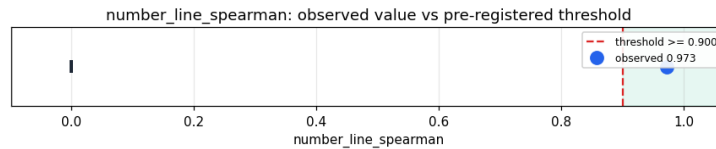


Figure 1: number_line_spearman – observed value with Wilson 95% CI.

3. Evidence

Pillar	Statistic	Value	95% CI	Library
magnitude_number_line	number_line_spearman	0.9727888184955529	(0.0000, 0.0000)	numpy 2.4

4. Pre-registration

- Registered at: 2026-05-24T03:38:21.369634+00:00
- Config hash: c30e07444773f0cae97e751a946035ef...
- Data hash: ea304b5707d0a4e1333352578cbd13de...

5. Signature

402a056c83ef33c6bf87afade626cb65fd3e40f11e1f474ca8f24d49ae07fdfd